BMEN 5251 | Midterm Project

Problem Statement:

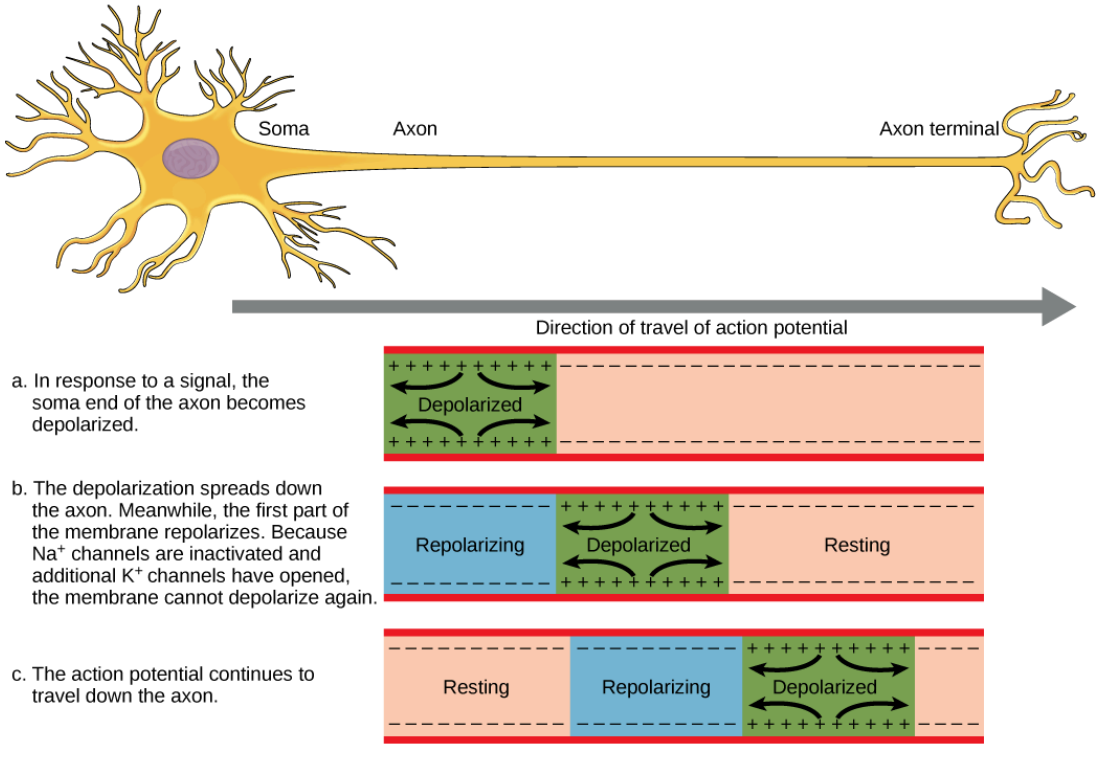
For this midterm project, I am modeling how action potentials propagate through an isolated active nerve fiber that follows Hodgkin-Huxley equations for its voltage gating and action potential when an external stimulus is introduced. This is to model how vagal nerve stimulation may react and can be used to predict conduction velocity of an action potential which ultimately would inform the user on how long of an external stimulus they would need to propagate across the entire fiber. The governing equation would be the cable equation, which defines how the voltage propagates over a distance:

$$C_m \frac{\partial V}{\partial t} = \frac{a}{2R_a}\frac{\partial^2 V}{\partial x^2} - (I_{Na} + I_K + I_L) + I_{stim}(x,\ t)$$

Some parameters to keep in mind is that the resting membrane potential is -65mV and the threshold to cause an action potential to propagate is -55mV. In this model, we want to stimulate our nerve fiber of length L=10cm at 3 cm with a sine waveform meant to emulate vagal nerve stimulation

Some other assumptions that were made are as follows:

- Nerve fiber is cylindrical in shape
- Nerve fiber has uniform potential across cross section to treat fiber as one dimension
- Nerve fiber is an isolated segment using a current clamp such that the ends act as ground. Thus $\frac{\partial V}{\partial x}(x = 0) = 0$ and $\frac{\partial V}{\partial x}(x = L) = 0$



Code Explanation (in fulfillment of Requirement 6 in the Midterm Requirements)

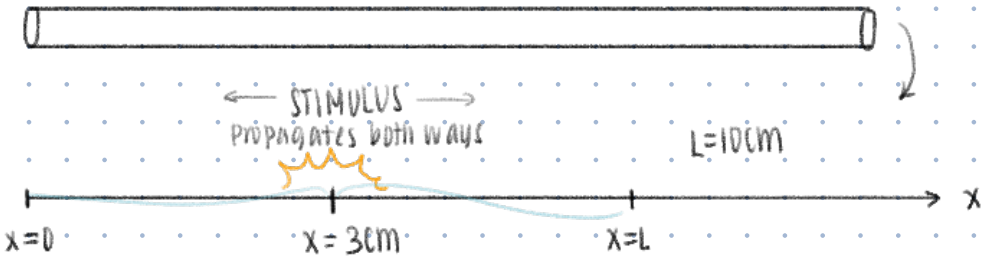
FEM Code:

- Establishes gating parameters and cellular membrane dynamics in correspondence of hodgkin huxley. These variables change with respect to the gating channel subunits being open. The ion channel is considered open when all subunits are open. The probability of the subunits being open is based on the transfer coefficients. The equations and dynamics of those can be found later on in this writeup with their corresponding differential equations. The code and gating derivations were taken from my previous BMEN5411 Homework 1.
- Utilizes FEM to solve the weak form of the cable equation
- For each element, the FEM code constructs the matrices for V, m, h, and n. It also sums up the ionic currents which is a constant displacement in the weak form

Midterm Project Call Script:

- Constructs the fiber and number of nodes
- Constructs external stimulus sine wave
- Plot stimulus waveform
- Set the initial conditions of V, m, h, and n.
- FEM to take the time derivative using ode15
- Plots voltage over time figure at fiber midpoint
- Plots voltage over fiber position at various time points

assume nerve fiber is cylindrical in shape (similar to a carotid). Nerve conductivities assume uniform potential across cross section to treat fiber as one dimensional



VP #05? for multiple vars?
VP #09? multiple global variables

$I_e(x,t) = I_{stim} + I_{ion}$
@ time t_1, add a stimulus at $x = L/2$. stimulus is caused by a sinusoidal wave form (time dependent), determine how stimulus affects extracellular potential as it traverses down the concentration gradient

assume isolated segment s.t. ends act as ground (i.e.) ends $x=0$ and $x=L$ have constant voltage potential (set to $-70\,mV$) $\Rightarrow$ current $I_e(t) = 0$ @ $x=0$ and $x=L$

B/C we want an isolated segment, **BOUNDARY CONDITIONS** are

$$\frac{\partial V}{\partial x}(0,t) = 0$$

$$\frac{\partial V}{\partial x}(L,t) = 0$$

Resting state: $V(x,0) = -65\,mV$ Threshold potential: $-55mV$

Cable equation (active)

$$\frac{a}{2R_a}\frac{\partial^2 V}{\partial x^2} = C_m\frac{\partial V}{\partial t} + \frac{V}{R_m} + I_{ion} + I_{stim}$$

The time range is $t \in [0, 20]ms$

Hodgkin-Huxley Block Gating

- Active 1-D cylindrical fiber of length L

*BMEN 5411 code adapted for HH gating dynamics

$$c_m \frac{\partial V}{\partial t} = \frac{a}{2R_a}\frac{\partial^2 V}{\partial x^2} - \left(I_{Na} + I_K + I_L\right) + I_{stim}(x,t) \rightarrow \text{cable eqn w/ HH dynamics}$$

With...

$$I_{Na} = G_{Na} m^3 h (V - V_{Na})$$

$$I_K = G_K n^4 (V - V_K)$$

$$I_L = G_L (V - V_L)$$

G = conductance (mS/cm²)
m, h, n are gating variables for channel subunits
V_i = reversal potential of ion

Gating variables satisfy

$$\frac{dm}{dt} = \overbrace{\alpha_m(V)}^{\text{fxn of V}}(1-m) - \overbrace{\beta_m(V)}^{\text{fxn of V}} m$$

$$\frac{dh}{dt} = \widetilde{\alpha_h(V)}(1-h) - \widetilde{\beta_h(V)} h$$

$$\frac{dn}{dt} = \widetilde{\alpha_n(V)}(1-n) - \widetilde{\beta_n(V)} n$$

m, n, h dimensionless. vary between [0, 1].
⇒represents prob. of subunit being open.

The transfer coefficients of channels being open/closed:

$$\alpha_m(V) = \frac{-0.1(V+35)}{\exp(-(V+35)/10) - 1}$$

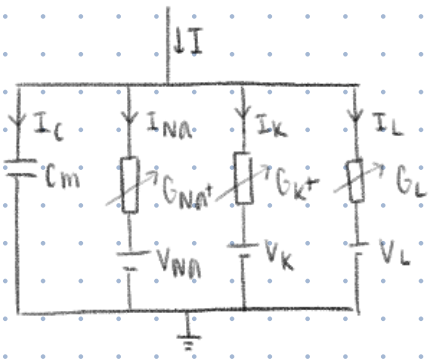
$$\beta_m(V) = 4\exp(-(V+60)/18)$$

$$\alpha_h(V) = 0.07\exp(-0.05(V+60))$$

$$\beta_h(V) = \frac{1}{\exp(-0.1(V+30)) + 1}$$

* Gating variables and transfer coefficients applied in HH rate functions in FEM solver fxn

equivalent circuit:



WEAK FORM of ODE:

$$C_m \frac{\partial V}{\partial t} = \frac{a}{2R_a} \frac{\partial^2 V}{\partial x^2} - \overbrace{\left(I_{Na} + I_K + I_L\right)}^{I_{ion}} + I_{stim}(x,t)$$

$$\int_0^L C_m \frac{\partial V}{\partial t} \varphi_i \, dx + \underbrace{\int_0^L \frac{a}{2R_a} \frac{\partial^2 V}{\partial x^2} \varphi_i}_{IBP} - \int_0^L I_{ion} \varphi_i \, dx = \int_0^L I_{stim}(x,t) \varphi_i \, dx$$

$$\int_0^L C_m \frac{\partial V}{\partial t} \varphi_i \, dx - \int_0^L \frac{a}{2R_a} \frac{\partial V}{\partial x} \frac{\partial \varphi_i}{\partial x} dx + \left[\frac{a}{2R_a} \frac{\partial V}{\partial x} \varphi_i\right]_0^L - \int_0^L I_{ion} \varphi_i \, dx = \int_0^L I_{stim}(x,t) \varphi_i \, dx$$

$$\boxed{\int_0^L C_m \frac{\partial V}{\partial t} \varphi_i \, dx - \int_0^L \frac{a}{2R_a} \frac{\partial V}{\partial x} \frac{\partial \varphi_i}{\partial x} dx - \int_0^L I_{ion} \varphi_i \, dx = \int_0^L I_{stim}(x,t) \varphi_i \, dx} \longrightarrow \text{WEAK FORM}$$

Mass M_{ij} | Stiffness A_{ij} | Leak K_{ij} | nonlinear term

Discrete $V(x,t) = \sum_{j=1}^{N} V_j \varphi_j(x)$

$$M_{ij} = \int_0^L \varphi_i \varphi_j \, dx$$

$$K_{ij} = \int_0^L a/2R_a \frac{d\varphi_i}{dx} \frac{d\varphi_j}{dx} dx$$

$$G_i = \int_0^L I_{ion}(V, m, h, n) \varphi_i \, dx$$

$$f_i(t) = \int_0^L I_{stim}(x,t) \varphi_i \, dx$$

$$\overbrace{C_m M \dot{V} + A V + G(V,m,h,n) = f(t)}^{b \text{ term}}$$

$C_m M\dot{V}$: time term | AV: diffusion | $G(V,m,h,n)$: ionic current | $f(t)$: stimulus (load term)

$$M\dot{V} + AV + b = 0$$

$$b = G(V,m,h,n) - f(t)$$

$$M\dot{V} = -AV - b$$

$$\frac{dV}{dt} = M \backslash (-AV - b)$$

→ input for ode15s to solve time derivative

$$\frac{dx}{dt} = f(x) \quad (x\text{-vector})$$

ode15s(fh, time range, initial)

function [dxdt] = MyFunc(x)

Parameter values from Literature

$G_{Na} = 40$ mS/cm²
$G_K = 35$ mS/cm²
$G_L = 0.3$ mS/cm²
} conductances of ion channels

$V_{Na} = 62$ mV
$V_K = -80$ mV
$V_L = -49$ mV
$C_m = 1$ μF/cm²
} Resting membrane potentials of each ion

L = 10 cm
a = 0.001 cm % axonal radius
R_a = 100 Ω·cm % internal axial resistance (Carpenter et. al., 1975)
} construct neuronal fiber

xStim = 3 % where stimulate on fiber
StimAmp = 500 % μA/cm² point source @ node
Stim Freq = 20 % kHz
Stim Dur = 2 % ms
} Stimulus
$I_{stim} = 500 \sin(40\pi t)$

instead of discretizing time and creating a residual, use ode15 to solve for V

$\frac{dx}{dt} = f(x)$ ← x-vector

ode15s(fh, time range, initial)
function [dxdt] = MyFunc(x)

```
close all; clear; clc;

L = 10; % cm
NN = 101;
xp = linspace(0, L, NN)';

% Fiber parameters
a = 1e-4; % cm
Ra = 100; % ohm*cm

% External stimulus
stim_location = 3.0; % cm
[~, idxStim] = min(abs(xp - stim_location));

% Stimulus Istim = 2sin(20pi*t)
f = 20e3; % 20 kHz
stim_duration = 10; % ms
pw = 0.25;  % ms pulse width

I_amp = 2e3; % 2 mA = 2000 µA

stim_fun = @(t) (t <= stim_duration) .* (mod(t, 2*pw) <= pw) .*(I_amp *
sin(2*pi*f*(t/1000)));

% Plot stimulus waveform
t_plot = linspace(0, 3, 5000);
stim_vals = arrayfun(stim_fun, t_plot);

figure;
plot(t_plot, stim_vals, 'LineWidth', 2);
xlabel('Time (ms)');
ylabel('Stimulus Current (µA/cm^2)');
title('20 kHz Pulsed Sinusoidal Stimulus (0.25 ms ON/OFF)');
grid on;

% Initial conditions
V0 = -65*ones(NN,1); %mV
m0 = 0.05*ones(NN,1);
h0 = 0.6*ones(NN,1);
n0 = 0.32*ones(NN,1);
U0 = [V0; m0; h0; n0];

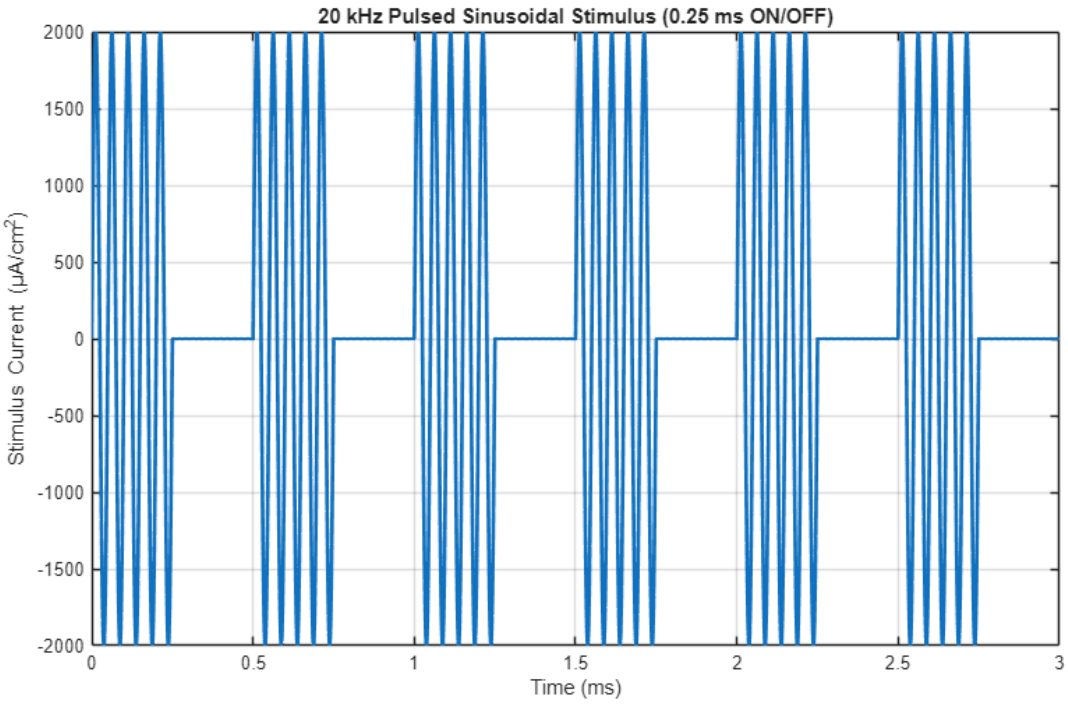

% FEM, take time derivative
[t,U] = ode15s(@(t,U) HHFiberModel(t,U,a,Ra,xp,stim_fun,idxStim), [0 20],
U0);

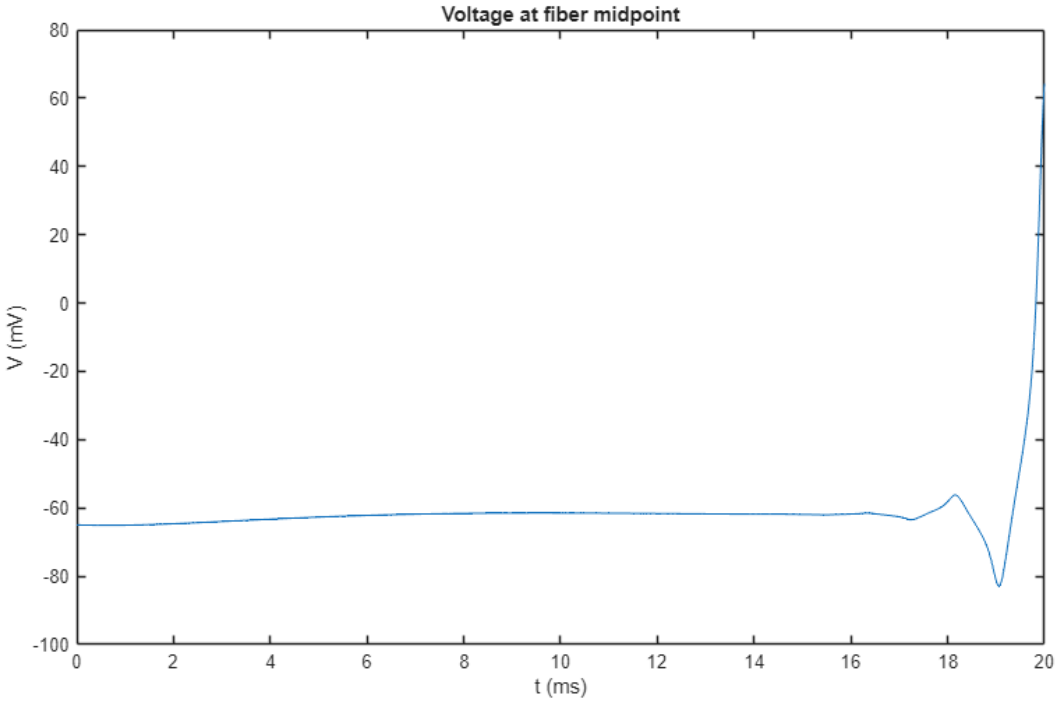
figure;
plot(t, U(:, round(NN/2)));
xlabel('t (ms)');
ylabel('V (mV)');
title('Voltage at fiber midpoint');
```

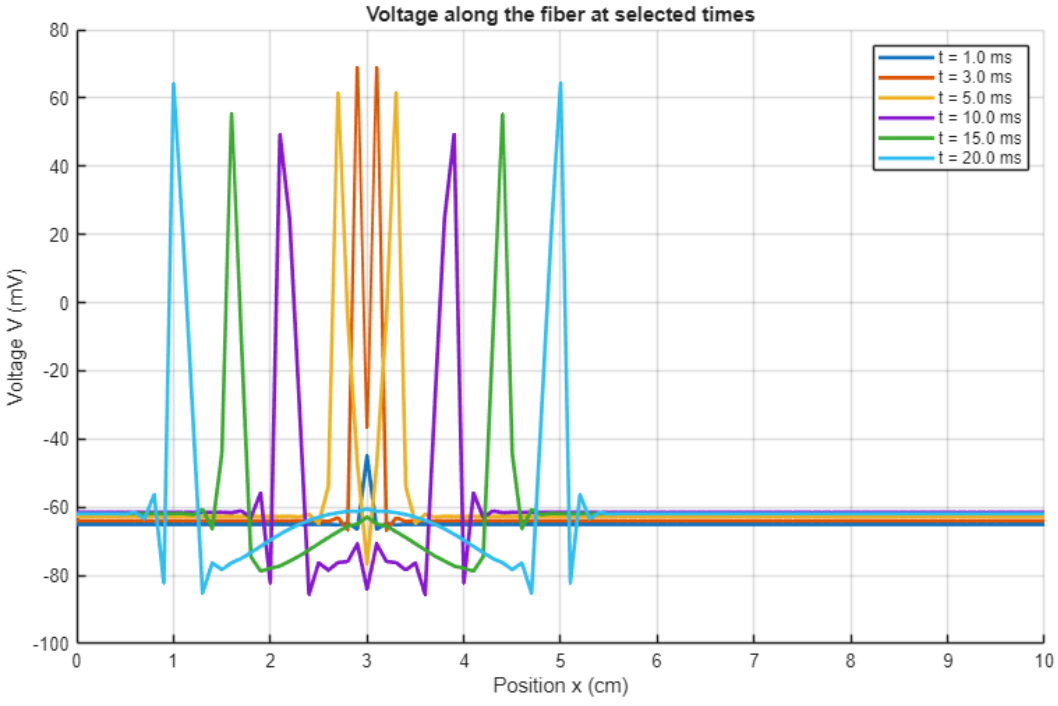
```
tspan = [1, 3, 5, 10, 15, 20]; % ms

figure; hold on;
for tt = tspan
    [~, idx] = min(abs(t - tt));
    Vplot = U(idx, 1:NN);
    plot(xp, Vplot, 'DisplayName', sprintf('t = %.1f ms', tt), 'LineWidth',
2);
end

xlabel('Position x (cm)');
ylabel('Voltage V (mV)');
title('Voltage along the fiber at selected times');
legend show;
grid on;
```







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```
function [dUdt, A, M, b] = HHFiberModel(t, U, a, Ra, xp, stim_fun, idxStim)

% Threshold for AP detection
Vth = -55; % mV

% Parameters
gNa = 120; %mS/cm^2
gK  = 36;  %mS/cm^2
gL  = 0.3; %mS/cm^2
ENa = 62; %mV
EK  = -80; %mV
EL  = -49; %mV
Cm  = 1; %uF/cm^2

NN = length(xp);
NE = NN - 1;

% Initialize unknown voltage and gating variable coefficients
V = U(1:NN);
m = U(NN+1:2*NN);
h = U(2*NN+1:3*NN);
n = U(3*NN+1:4*NN);

% Threshold detection
spike_flag = V > Vth;


% Gauss points
rs3 = 1/sqrt(3);
gp  = rs3.*[-1, 1];
gw  = [1, 1];

f = [(1-gp)/2; (1+gp)/2];
dfds = [-0.5, 0.5];

% Connectivity
T = [(1:NE)', (2:NE+1)'];

% Allocate matrices
M = zeros(NN,NN);
A = zeros(NN,NN);
b = zeros(NN,1);

% Diffusion coefficient
D = a/(2*Ra);

% Time-dependent external stimulus
b_ext = zeros(NN,1);
b_ext(idxStim) = stim_fun(t);

% Element Loop
for e = 1:NE
```

```
    xe = [xp(T(e,1)); xp(T(e,2))];
    J = dfds(1)*xe(1) + dfds(2)*xe(2);
    dfdx = dfds / J;

    for g = 1:2
        Vg = f(1,g)*V(T(e,1)) + f(2,g)*V(T(e,2));
        mg = f(1,g)*m(T(e,1)) + f(2,g)*m(T(e,2));
        hg = f(1,g)*h(T(e,1)) + f(2,g)*h(T(e,2));
        ng = f(1,g)*n(T(e,1)) + f(2,g)*n(T(e,2));

        % HH ionic currents
        INa = gNa * mg^3 * hg * (Vg - ENa);
        IK  = gK  * ng^4 * (Vg - EK);
        IL  = gL  * (Vg - EL);
        Iion = INa + IK + IL; %Iion is the sum of the ion channel currents
Na+, K+, and leak channels (Cl-, Ca2+)

        for il = 1:2
            ig = T(e,il);

            for jl = 1:2
                jg = T(e,jl);

                % Mass matrix
                M(ig,jg) = M(ig,jg) + Cm * f(il,g)*f(jl,g) * gw(g) * J;

                % Stiffness matrix
                A(ig,jg) = A(ig,jg) + D * dfdx(il)*dfdx(jl) * gw(g) * J;
            end

            % Load vector
            b(ig) = b(ig) + (Iion - b_ext(ig)) * f(il,g) * gw(g) * J;
        end
    end
end

% Gating variable ODEs | derived from BMEN5411 homework. Original code is
% available upon request for further review
[am, bm, ah, bh, an, bn] = hh_rates_local(V);

dmdt = am.*(1-m) - bm.*m;
dhdt = ah.*(1-h) - bh.*h;
dndt = an.*(1-n) - bn.*n;

% Voltage ODE: M dV/dt = -A*V - b
dVdt = M \ (-A*V - b);

% Full ODE vector
dUdt = [dVdt; dmdt; dhdt; dndt];

end
```

```
% HH rate functions
%  Adapted from BMEN5411 homework code that I wrote last
% semester. Full code / homework assignment is available upon request to be
% reviewed, but I left it out of the official midterm submission to
% increase clarity.

function [am, bm, ah, bh, an, bn] = hh_rates_local(V)

am = (-0.1*(V+35)) ./ (exp(-(V+35)/10) - 1);
bm = 4.0 * exp(-(V+60)/18);

ah = 0.07 * exp(-0.05*(V+60));
bh = 1 ./ (exp(-0.1*(V+30)) + 1);

an = (-0.01*(V+50)) ./ (exp(-0.1*(V+50)) - 1);
bn = 0.125 * exp(-(V+60)/80);

% Avoid singularities
am(abs(V+35) < 1e-8) = 1.0;
an(abs(V+50) < 1e-8) = 0.1;

end
```

```
Not enough input arguments.

Error in HHFiberModel (line 15)
NN = length(xp);
             ^^
```

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